

Muhammet Ali Yavuz

Naval Architect · CFD Engineer · Software Architect

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Portfolio: yavuzma.github.io

EDUCATION

BSc Shipbuilding & Ocean Engineering Istanbul Technical University

GPA: 3.25 / 4.0 · 5-Year Program (Including 1-Year English Preparatory)

ATHENS Exchange Programme Delft University of Technology

TUD14: Computer Aided Analysis & Optimization of Sustainable Electrical Power Systems ·
Local Mark: 9/10 · ATHENS Network (athensnetwork.eu)

EXPERIENCE

Researcher

CENTEC, Instituto Superior Técnico, University of Lisbon

- Turbulence modelling.
- Hydroacoustic characterization of a propeller operating in a wake.

Undergraduate Researcher ITUKAT

Conducted under confidentiality; descriptions focus on methodology.

- Force and moment analysis in CFD with experimental (EFD) validation.
- Control and automation algorithms for engineering applications.

Curious Student

ITU Ata Nutku Towing Tank

- Resistance, propulsion, and seakeeping experiments; towing-tank testing procedures.

Production Engineer Anadolu Shipyard

Military naval platforms; combat & navigation systems under confidentiality.

- Gas turbine (GE LM2500) foundation mounting and micron-level coupling alignment.
- CPP blade-bolt elongation-controlled assembly, propeller shaft line alignment, and NDT.

Production Engineer Besiktas Shipyard

- Main engine, hydraulic and generator maintenance; compressor & ballast pump overhaul.
- Welding, pipe installation, and surface coating experience.

Mechanical Engineer MENSA MAKINA

- Designed and installed a complete ventilation system for a sheet-metal processing line.
- Fabricated sheet-metal components: pipes, flanges, and air filters.

KEY PROJECTS

TCSG FISILTI - Hybrid Coast Guard Vessel

69.5 m hybrid vessel with Counter-Rotating Azipod propulsion, 24 kn max speed, 63 crew capacity.

KCS Hull with Rudder - CFD Resistance

Unsteady RANS + VOF in STAR-CCM+; 5 cases × 3 mesh levels; 0.22% deviation from EFD.

DARPA SUBOFF - Appendage Resistance

AFF1 / AFF3 / AFF8 configurations, 0.5% deviation, ITTC uncertainty assessment.

Offshore Platform - Design & Analysis

70 × 40 m operational deck; Rhinoceros 3D + Maxsurf stability + ANSYS Fluent CFD.

Alicat Mass Flow Controller - Control Interface

Python desktop app with AsyncIO serial I/O, real-time monitoring & CSV logging.

SKILLS

Analysis: OpenFOAM, ANSYS Fluent, STAR-CCM+, Maxsurf

Design: SolidWorks, Rhinoceros 3D, AutoCAD

Programming: C / C++, Python, MATLAB, Arduino

LANGUAGES

English - Advanced

German - Intermediate

Turkish - Native

CERTIFICATIONS

· ATHENS Network - TU Delft (Mar 2026)
BN22 - JYBT - 79WV - HDCV

· Submarine Technology & Design
GMO · 26 h · Oct-Nov 2024

· SolidWorks CSWA

· AutoCAD Certificate

· Python - freeCodeCamp

· Arduino - Udemy

· TELC Deutsch B1

MEMBERSHIPS

· SNAME - Student Member
Society of Naval Architects & Marine Engineers

INTERESTS

Hydrodynamics Ship Design CFD

Hydroacoustics Marine Systems

Defense Tech Autonomous Systems

Propeller Design AI

ACTIVITIES

ITU Shipbuilding Engineering Club

ITU SAVTEK - Defense Technologies